Aditya Rana

Email: adityaranarana2006@gmail.com
Mobile: +91-9927948621
LinkedIn: linkedin.com/in/adityarana05
GitHub: github.com/adityarana121

Summary

Computer Science and Engineering student with hands-on experience in **Artificial Intelligence (AI)**, **Machine Learning (ML)**, **Deep Learning**, and **Frontend Development**. Expertise in **Python, C++, JavaScript, SQL**, and Specialized in building real-world AI applications using frameworks like **TensorFlow, OpenCV**, and **React.js**. Proficient in designing and optimizing algorithms, data structures, and performance-tuned solutions.

Education

Graphic Era Deemed to be University **2022 –2026**
Bachelor of Technology in Computer Science and Engineering *Dehradun,Uttarakhand, INDIA*

Projects

Automatic Number Plate Recognition (ANPR) | *OpenCV, Computer Vision and AI* |

- Implemented **YOLOv8 for real-time license plate detection** with an inference speed of **~30 FPS**, enabling seamless operation in live video streams.
- **Secured 95% accuracy** in identifying and extracting license plate information from images, ensuring high reliability.
- **Enhanced vehicle identification and tracking** by integrating the ANPR system with **real-time surveillance and traffic monitoring**, processing **up to 50 video feeds simultaneously**
- **Optimized model performance** by reducing inference time by **25%** through **TensorRT** acceleration and quantization techniques.

Age and Gender Prediction | *CNN, TensorFlow and Deep Learning* |

- Developed and fine-tuned a convolutional neural network model with exceptional efficiency; achieved superior feature recognition by **processing over 50,000 images** while reducing training time by 30 hours through defined data handling techniques.
- Trained and evaluated the model on a large **facial dataset**, achieving an impressive **97% accuracy** in age and gender prediction.
- **Maximized computational efficiency** by leveraging GPU acceleration (CUDA, TensorRT), reducing model **training time by 40%** and boosting inference speed to **100+ FPS** for real-time applications.
- Incorporated **advanced data augmentation techniques**, including **lighting variations, facial occlusions**, and **pose variations**, to improve model generalization and **reduce overfitting by 15%** .

Random Meme Generator | *HTML, CSS, JavaScript and React* |

- Engineered an interactive web application using React, HTML, CSS, and JavaScript; generated over **5,000 unique** memes on demand while maintaining user engagement levels above industry standards through a seamless interface.
- **Integrated third-party REST APIs** to deliver real-time meme content in less than **500 milliseconds** per request; eliminated load times resulted in improved user satisfaction scores based on feedback from targeted usability tests.
- Crafted a fully responsive, user-focused interface with custom CSS enhancements that enhanced user experience across mobile, tablet, and desktop devices; improved site **accessibility ratings by 30%** through adaptive design features.
- Implemented **asynchronous rendering and lazy loading** to enhance app performance, achieving **~90% reduction in initial load time** while maintaining a smooth user experience.

Technical skills

- **Languages :** C, C++, Java, JavaScript, SQL(Structured Query Language), Python, PHP.
- **Database :** MySQL(Oracle), MongoDB.
- **Web Technologies :** HTML/CSS, Bootstrap CSS, React.js.
- **Softwares :** Git, VS Code, Code::Blocks, Eclipse, Jupyter Notebook, Cisco Packet Tracer.

Certificates

- Microsoft Certified: Azure AI Fundamentals.
- Unsupervised Learning, Recommenders, Reinforcement Learning.